



## **Java Institute for Advanced Technology**

# **SOFTWARE ENGINEERING II (SOFTWARE TESTING, QUALITY ASSURANCE AND MAINTENANCE) HF2W 04**

**HF2W 04/AS/05**

NAME: Maalik Hassan

BATCH: HYDRA

NIC: 200208100390

BRANCH: COLOMBO



|                                                            |              |
|------------------------------------------------------------|--------------|
| <b>Scenario</b> Banking Application Login and Transactions |              |
| <b>Student Name</b>                                        | <b>NIC</b>   |
| Maalik Hassan                                              | 200208100390 |

|                              |                                                                                                                                                                                                                                  |                         |                |     |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|-----|
| <b>Test Case Description</b> | Testing the login functionality of a banking application, including scenarios for successful login, unsuccessful login, password recovery, account lockout, transferring funds between accounts and viewing transaction history. |                         |                |     |
| <b>Created By Name / NIC</b> | Maalik Hassan                                                                                                                                                                                                                    |                         |                |     |
|                              | <b>Reviewed By</b>                                                                                                                                                                                                               | Mr. Prasanna Senanayaka | <b>Version</b> | 1.0 |

|                      |               |                    |            |                                 |     |
|----------------------|---------------|--------------------|------------|---------------------------------|-----|
| <b>Tester's Name</b> | Maalik Hassan | <b>Date Tested</b> | 17/12/2024 | <b>Test Case Pass Overall %</b> | 90% |
|----------------------|---------------|--------------------|------------|---------------------------------|-----|

| # | Prerequisites:                                                     |
|---|--------------------------------------------------------------------|
| 1 | Access to internet and the Banking application                     |
| 2 | Registered user account with active credentials.                   |
| 3 | User has sufficient funds in the account for testing transactions. |

|                       |                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------|
| <b>Test Scenarios</b> | Verify on entering valid userid and password, the customer can login                                  |
|                       | Verify on entering valid receipient id and having sufficient balance, the customer can transfer funds |
|                       | Verify on entering valid userid and password, the customer can view transaction history               |

| step # | Test steps                                         |
|--------|----------------------------------------------------|
| 1      | Login with valid credentials.                      |
| 2      | Enter an invalid email format.                     |
| 3      | Attempt login with an inactive account.            |
| 4      | Enter the wrong password.                          |
| 5      | Leave both fields blank.                           |
| 6      | Leave the password field blank.                    |
| 7      | Leave the email field blank.                       |
| 8      | Enter an invalid email format.                     |
| 9      | Log in on multiple devices.                        |
| 10     | Fail login five times consecutively.               |
| 11     | Attempt login with a locked account.               |
| 12     | Initiate password recovery with an invalid email.  |
| 13     | Initiate password recovery with a valid email.     |
| 14     | Change password with an invalid old password.      |
| 15     | Reset password to a weak password.                 |
| 16     | Reset password to a strong new password.           |
| 17     | Change password with valid credentials.            |
| 18     | Attempt fund transfer to an invalid account.       |
| 19     | Attempt fund transfer with insufficient balance.   |
| 20     | Attempt to transfer zero funds.                    |
| 21     | Exceed the daily transfer limit.                   |
| 22     | Transfer funds to a valid account.                 |
| 23     | Transfer funds with currency conversion.           |
| 24     | View account balance.                              |
| 25     | View transaction history with records.             |
| 26     | View transaction history with no records.          |
| 27     | Log out successfully.                              |
| 28     | Test session timeout.                              |
| 29     | Enable two-factor authentication (2FA).            |
| 30     | Attempt to bypass 2FA from an unauthorized device. |

### Testing for Login:

| Test Case ID | Test Case                                                           | Expected Results                                                                | Actual Results        | Pass / Fail / Not executed / Suspended | Comment                               |
|--------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------|----------------------------------------|---------------------------------------|
| BU_001       | Userid = user@gmail.com;<br>Pass = df12@434c                        | Customer is logged in                                                           | As Expected           | Pass                                   |                                       |
| BU_002       | Userid = @gmail.com;<br>Pass = df12@434c                            | Error Display 'Invalid Email'                                                   | As Expected           | Pass                                   |                                       |
| BU_003       | Userid = hello@gmail.com; Pass = df12@434c                          | Error 'No active account found'                                                 | Customer is logged in | Fail                                   | System allowed inactive account login |
| BU_004       | Userid = user@gmail.com;<br>Pass = wrongpass123                     | Error 'Wrong password'                                                          | As Expected           | Pass                                   |                                       |
| BU_005       | Userid = blank; Pass = blank                                        | Error 'Fields cannot be blank'                                                  | As Expected           | Pass                                   |                                       |
| BU_006       | Userid = user@gmail.com;<br>Pass = blank                            | Error 'Password cannot be blank'                                                | As Expected           | Pass                                   |                                       |
| BU_007       | Userid = blank; Pass = df12@434c                                    | Error 'Email cannot be blank'                                                   | As Expected           | Pass                                   |                                       |
| BU_008       | Userid = invalid@format;<br>Pass = df12@434c                        | Error 'Invalid email format'                                                    | As Expected           | Pass                                   |                                       |
| BU_009       | Userid = user@gmail.com;<br>Pass = df12@434c<br>(Multiple Sessions) | Customer is logged in already in different device<br>(Multiple sessions active) | As Expected           | Pass                                   |                                       |

### Testing lockout and password recovery/ change:

|        |                                                                             |                                         |                             |      |                                           |
|--------|-----------------------------------------------------------------------------|-----------------------------------------|-----------------------------|------|-------------------------------------------|
| BU_010 | Userid = user@gmail.com;<br>Attempt Login 5 Times<br>(Lockout)              | Error 'Account locked after 5 attempts' | As Expected                 | Pass |                                           |
| BU_011 | Userid = user@gmail.com;<br>Pass = df12@434c<br>(Account Locked)            | Error 'Account locked, contact support' | As Expected                 | Pass |                                           |
| BU_012 | Userid = user@gmail.com;<br>Initiate Password Recovery<br>(Invalid Email)   | Error 'Invalid email for recovery'      | As Expected                 | Pass |                                           |
| BU_013 | Userid = user@gmail.com;<br>Pass = df12@434c; Initiate<br>Password Recovery | Password recovery email sent            | As Expected                 | Pass |                                           |
| BU_014 | Userid = user@gmail.com;<br>Change Password = Invalid<br>Old, Valid New     | Error 'Invalid old password'            | Error displayed differently | Fail | UI could not display proper error message |
| BU_015 | Userid = recovery@gmail.com;<br>Reset Password = weakpass                   | Error 'Password too weak'               | As Expected                 | Pass |                                           |
| BU_016 | Userid = recovery@gmail.com;<br>Reset Password = df12@newpass               | Password reset successful               | As Expected                 | Pass |                                           |
| BU_017 | Userid = user@gmail.com;<br>Change Password = Valid<br>Old, Valid New       | Password changed successfully           | As Expected                 | Pass |                                           |

### Testing for Transferring Funds:

|        |                                                                                        |                                                          |                             |      |                                          |
|--------|----------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------|------|------------------------------------------|
| BU_018 | Userid = user@gmail.com;<br>Transfer Funds = Invalid<br>Account, Valid Amount          | Error 'Invalid recipient<br>account'                     | As Expected                 | Pass |                                          |
| BU_019 | Userid = user@gmail.com;<br>Transfer Funds = Valid<br>Account, Insufficient<br>Balance | Error 'Insufficient balance'                             | Error displayed differently | Fail | Insufficient<br>balance error<br>unclear |
| BU_020 | Userid = user@gmail.com;<br>Transfer Funds = Valid<br>Account, Zero Amount             | Error 'Cannot transfer zero<br>amount'                   | As Expected                 | Pass |                                          |
| BU_021 | Userid = user@gmail.com;<br>Transfer Funds = Valid<br>Account, Exceeds Daily<br>Limit  | Error 'Daily transfer limit<br>exceeded'                 | As Expected                 | Pass |                                          |
| BU_022 | Userid = user@gmail.com;<br>Transfer Funds = Valid<br>Account, Valid Amount            | Funds transferred<br>successfully                        | As Expected                 | Pass |                                          |
| BU_023 | Userid = user@gmail.com;<br>Transfer Funds = Currency<br>Conversion                    | Funds transferred with<br>currency conversion<br>applied | As Expected                 | Pass |                                          |

### Testing viewing of transaction history and 2FA authorization:

|        |                                                                      |                                           |             |      |  |
|--------|----------------------------------------------------------------------|-------------------------------------------|-------------|------|--|
| BU_024 | Userid = user@gmail.com;<br>View Account Balance                     | Account balance displayed                 | As Expected | Pass |  |
| BU_025 | Userid = user@gmail.com;<br>View Transaction History                 | Transaction history<br>displayed          | As Expected | Pass |  |
| BU_026 | Userid = user@gmail.com;<br>View Transaction History<br>(No Records) | Message 'No transaction<br>records found' | As Expected | Pass |  |
| BU_027 | Userid = user@gmail.com;<br>Logout                                   | Customer logged out<br>successfully       | As Expected | Pass |  |
| BU_028 | Userid = user@gmail.com;<br>Timeout Session After<br>Inactivity      | Session expired and user<br>logged out    | As Expected | Pass |  |
| BU_029 | Userid = user@gmail.com;<br>Enable 2FA                               | 2FA enabled successfully                  | As Expected | Pass |  |
| BU_030 | Userid = user@gmail.com;<br>Bypass 2FA (Unauthorized<br>Device)      | Error 'Unauthorized device<br>for 2FA'    | As Expected | Pass |  |